

Improving Zero Shot Forecasting with Support Series and Multivariate Sensor Context

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Abstract—Zero shot time series forecasting models attempt to predict future values of the input sequence even when the input is an instance from a distribution that has not been seen during training. Such models generalize the forecast patterns observed from other, but similar, distributions during training. Such models are applicable in cases where there is little historical data in one area to train a domain-specific model, but there are instances from other areas. This work proposes an architecture consisting of a Transformer encoder arranged in a Siamese twin configuration followed by an LSTM decoder stage. The key idea is that a portion of the training data is retained as a support dataset after training to be used during inference. We show that when the instances can be augmented with data from correlated sources, the forecast accuracy can be improved as compared to the case where only univariate instances are used. The architecture is quantitatively evaluated by forecasting vehicular traffic in California at different time horizons (4 to 48 hours). The model is trained with traffic flow data collected in one district of California and then tested by forecasting traffic in other districts. Performance is compared with three baseline models, including a single transformer model and LSTM-based models. The results show that adding multivariate sensor context improves performance across all four model families. The largest reduction in forecast error occurs in models that combine a support dataset with context from correlated sensors.

Index Terms—Time Series Forecasting, Zero-shot Forecasting, Traffic Forecasting, LSTM, Attention Mechanism, Transfer Learning, Sequence Modeling, Support-Based Forecasting, Deep Learning for Time Series, Transformer Networks, Siamese Networks.

I. INTRODUCTION

Time-series forecasting is the task of predicting future values given a sequence of past measurements. Time-series forecasting has several real world applications, including in vehicular traffic flow prediction [1]. In recent years, machine learning methods to time series forecasting have outperformed other approaches in terms of prediction accuracy. However, one disadvantage of machine learning methods is the need for historical time series data from the same domain (“in-distribution data”) to train the model. Few shot time-series forecasting methods are being developed to address this requirement. Few shot methods use datasets from several different related domains for the bulk of the training and require only a relatively small number of samples from the prediction domain before making a forecast. Zero shot

forecasting is the extreme case where the model makes a prediction using only a single instance of in-distribution data, i.e., all model building is completed during the pre-training stage.

Few shot time-series forecasting is an active area of research with most new methods using deep learning architectures [2]. In previous work [3], we presented two architectures for zero shot time series forecasting that used transformer models arranged in a Siamese network configuration. We evaluated these architectures for the purpose of vehicular traffic flow prediction in California and showed that one model achieves a Mean Absolute Error (MAE) that is 8.3% lower than the baseline LSTM with attention mechanism model.

In this paper, we extend these architectures to exploit correlations between data sources to improve prediction accuracy. In many applications, the distinction between in-distribution and out-of-distribution data is not sharp. Instead, some of the out-of-distribution data sources can produce time-series that are correlated with the time-series to be used for forecasting. For instance, in the case of traffic flow prediction, we can expect that the traffic as measured in nearby locations is related to each other. This correlation may not be linear and could have complex dependencies but transformer architectures are able to learn such complex patterns.

Our architecture can be used when a domain-dependent measure of closeness between the sources of the time-series is available. This enables the time-series to be augmented with sequences from close sources and enables the model to learn the interdependencies. The architecture is still based on transformer models arranged in a Siamese network configuration. The models incorporate a *support series* and a *difference vector*. The support series is the one instance in the training data that is most similar to the query series. The difference vector is the element-wise differences between the transformer encodings of the support and query series. The data augmentation may be considered a multivariate extension of the previously introduced model.

Evaluation with vehicular traffic data shows that this approach leads to increased prediction accuracy, i.e., the augmentation to multivariate time-series enables more efficient

use of the limited data available for training in zero shot settings. We used the road sensor measurements collected by the California Department of Transportation’s (Caltrans) Performance Measurement System Data Source (PeMS) for detailed evaluation. The models were trained with traffic flow data collected in one California district and then evaluated by forecasting traffic at a different district at different prediction horizons (4 to 48 hours).

Our contributions are as follows:

- We introduce the task of zero shot time-series forecasting in the presence of a domain-dependent similarity metric between data sources
- We present a transformer-based neural network architecture for the above task
- We evaluate the model on a large publicly available vehicular traffic flow dataset and show that data augmentation increases prediction accuracy over all forecasting horizons.

II. RELATED WORK

Zero shot and few shot time-series forecasting is an area of active research. As in other areas of machine learning, neural network-based approaches dominate current methods [2]. One set of such approaches explores more efficient use of the limited in-distribution data that is available for training. Orozco and Roberts [4] propose a recurrent neural network (RNN) model that learns a shared latent embedding across multiple quantized time series for both zero shot and few shot forecasting.

Taga et al. [5] propose a transformer-based architecture and a training scheme for multivariate time-series forecasting. The self-attention mechanism in transformer models is permutation invariant and so positional encodings have to be used with time series data [6]. Tran and Panangadan [7] use Siamese neural networks with LSTM units for few shot forecasting. Their model learns to predict a difference vector between time-series pairs rather than directly forecasting future values.

In prior work [3], we proposed two architectures designed for zero shot time series forecasting. Both architectures use transformer models arranged in a Siamese network configuration. In this paper, we extend these architectures for situations where it is possible to define a domain-dependent measure of closeness between the sources of the in-distribution datasets. This enables more efficient use of the limited data available for training in zero shot settings.

In contrast, pre-trained time series foundation models are designed to forecast with no training with in-distribution data and instead rely on in-context learning [8]. Dooley et al. [9] develop a forecasting model that is trained purely on a synthetic data distribution. Experimental results on selected datasets show improved performance over models trained with domain data. Chronos and Chronos-2 [10] use transformer-based language model architectures and group attention mechanism. The models are trained on a large collection of publicly available datasets and synthetic datasets.

These models have been shown to have comparable performance to models that were trained for specific datasets. However, Toner et al. [11] have found that foundation time series models can fail to generalize on certain types of data.

III. APPROACH

The problem of zero-shot forecasting can be stated as:

$$\hat{\mathbf{y}}_{t+1:t+H} = f_{\theta}(\mathbf{x}_{t-W+1:t}^{\text{tar}}, \mathbf{C}_{t-W+1:t}, \mathcal{S}), \quad (1)$$

where $\mathbf{x}_{t-W+1:t}^{\text{tar}} \in \mathbb{R}^W$ denotes the history of the target series, $\mathbf{C}_{t-W+1:t} \in \mathbb{R}^{W \times (D-1)}$ denotes the accompanying contextual series observed over the same window, and the combined multivariate input can therefore be written as $\mathbf{X}_{t-W+1:t} = [\mathbf{x}_{t-W+1:t}^{\text{tar}}, \mathbf{C}_{t-W+1:t}] \in \mathbb{R}^{W \times D}$. The term $\hat{\mathbf{y}}_{t+1:t+H} \in \mathbb{R}^H$ is the forecast for the target variable over the horizon H , f_{θ} is the forecasting model, and $\mathcal{S}$ is an optional support set used in zero-shot settings.

We describe four zero shot time-series forecasting models. The first three are used as baselines for performance comparison, while the fourth is our proposed architecture. Each model takes as input a multivariate time-series of length W , $\mathbf{X}_{t-W+1:t}$ (also called the query series), where each time step is represented by a D -dimensional feature vector, and produces a forecast of length H for the target variable $\hat{\mathbf{y}}_{t+1:t+H}$. The models are trained on a dataset Z of L instances, $\mathbf{Z}_{t-W+1:t}^{(j)}$, $j = 1, 2, \dots, L$. Note that the training instances ($\mathbf{Z}^{(j)}$) are drawn from a different distribution than the query series ($\mathbf{X}$), so the prediction for $\mathbf{X}$ is produced in a zero-shot setting.

In order to construct multivariate forecasting examples, each target series is augmented with a set of related contextual series drawn from the same data set, such that every time step is represented by a D -dimensional feature vector containing the target variable together with auxiliary variables or neighboring series. The specific choice of contextual series is dataset dependent and may be based on spatial proximity, correlation, domain knowledge, or similar criteria. The model receives the first W multivariate steps at input, while the supervision signal consists only of the future trajectory of the target variable over the next H steps.

In addition to the multivariate formulation, we consider a univariate version of the task as a comparison baseline. In this setting, the input to the models consists of the history of the target series without any contextual series, so $D = 1$. All other aspects of the pipeline, including the forecasting horizon, support set construction, model architectures, and optimization procedure remain the same.

A. LSTM with Attention

A Long short-term memory (LSTM) architecture with an attention mechanism is used as the baseline model as LSTMs have been shown to work particularly well with sequence data [12]. The attention mechanism allows the model to understand and dynamically weigh the importance of different time steps in the input sequence, enabling it

to capture patterns and dependencies. Although the input is multivariate, the model forecasts only the values of the target variable.

B. Transformer

The Transformer is included as a direct forecasting baseline. In this model, each multivariate input time step is first projected into a latent embedding space and augmented with positional encodings so that temporal order is preserved. The resulting sequence is then processed by a stack of Transformer encoder layers with multi-head self-attention, allowing the model to capture long-range dependencies across the entire input window. After encoding, the sequence is flattened and projected to the forecast horizon, and a final linear layer maps each future representation to the predicted values of the target variable. Like the LSTM with Attention baseline, this model uses only the query series at inference time and does not rely on support examples.

C. Difference LSTM with Attention

The *Difference LSTM with Attention (DLA)* is an enhanced version of the LSTM with Attention model for zero shot forecasting. DLA incorporates two additional components: a *support series* and a *difference vector*. A randomly sampled portion of the training dataset Z , called the *Support dataset* ($SV, S \subset Z$), is set aside for use during forecasting. For a given query series, the support series is selected from S as the support instance whose past window is most similar to the query past. Support selection can be based on any similarity measure appropriate for the dataset; in this work, we use the Euclidean distance in the multivariate input space. Specifically, the support series of input $\mathbf{X}$, $S(\mathbf{X})$, is defined as:

$$S(\mathbf{X}) = \mathbf{Z}^*, \quad d(\mathbf{X}, \mathbf{Z}^*) = \min_{\mathbf{Z} \in S} d(\mathbf{X}, \mathbf{Z})$$

$$d(\mathbf{X}, \mathbf{Z}) = \sum_{t=1}^W \sum_{d=1}^D (X_{t,d} - Z_{t,d})^2$$

As the query and support vectors are drawn from different distributions, the model also incorporates a *difference vector*. The difference vector is the vector of element-wise differences between the query and its support series. The difference vector of input $\mathbf{X}$, $DV(\mathbf{X})$, is defined as

$$DV(\mathbf{X}) = \mathbf{X} - S(\mathbf{X})$$

The query past is processed by an LSTM with Attention backbone, while both the learned representations of $DV(\mathbf{X})$ and the support future are added to the hidden states before attention is applied.

D. Siamese Transformer with Augmented Inputs

Our proposed architecture uses a shared Transformer encoder to capture complex temporal dependencies through multi-head self-attention. The training data is divided into a support series dataset and a training dataset as described

previously in the Difference LSTM with Attention architecture. Each instance in the training and support datasets is augmented with its closest k neighbors using some domain-specific criterion (in our experiments, we used geographic location), resulting in a multivariate dataset. However, unlike the Difference LSTM architecture, the Siamese Transformer architecture computes the difference vector in Transformer-encoded space (rather than input space) to encode the difference vector in a more meaningful manner. Specifically, the Siamese transformer difference vector of input $\mathbf{X}$, $DV'(\mathbf{X})$, is defined as

$$DV'(\mathbf{X}) = TRANS(\mathbf{X}) - TRANS(S(\mathbf{X}))$$

where $TRANS(\cdot)$ is the output from the Transformer stage of the architecture (see Figure 1).

The model receives the query past, the support past, and the support future. The shared Transformer encoder is applied to all three inputs to produce latent representations. The model then computes the element-wise difference between the query-past embedding and the support-past embedding, and combines this difference with the embedding of the support future. This combined support-aware representation is projected into the hidden space of an LSTM with Attention backbone, which is responsible for generating the final forecast of the target variable.

All four models are trained using Mean Absolute Error (MAE) as the loss function, which is commonly used in time-series tasks due to its robustness to outliers and interpretability [13].

$$MAE = \frac{1}{H} \sum_{t=1}^H |y_t - \hat{y}_t|, \quad (2)$$

where y_t is the true value of the target variable at forecast step t , and $\hat{y}_t$ is the predicted value at that step. Minimizing the MAE encourages the models to produce forecasts that more closely align with the ground truth values across the forecast horizon while treating all errors equally, regardless of magnitude.

IV. DATASETS

1) *Dataset Creation*: Our training, validation, and testing datasets are from California Department of Transportation (Caltrans) Performance Measurement System Data Source (PeMS) [14]. This data is collected from nearly 40,000 individual detectors across the state of California in 5-minute intervals.

2) *Data Pre-Processing*: We retained only the following features – the timestamp, station identifier (a unique identifier for each station), and total flow at each detector. To reduce data noise and enhance forecasting reliability, we down sampled the data from its original 5-minute intervals into 30-minute intervals. Our training dataset consists of data from Orange County, California (Caltrans District 12), covering the full year of 2023. The testing dataset includes data from the Bay Area and Oakland (Caltrans District 4) for the first three months of 2023. This intentional geographic separation

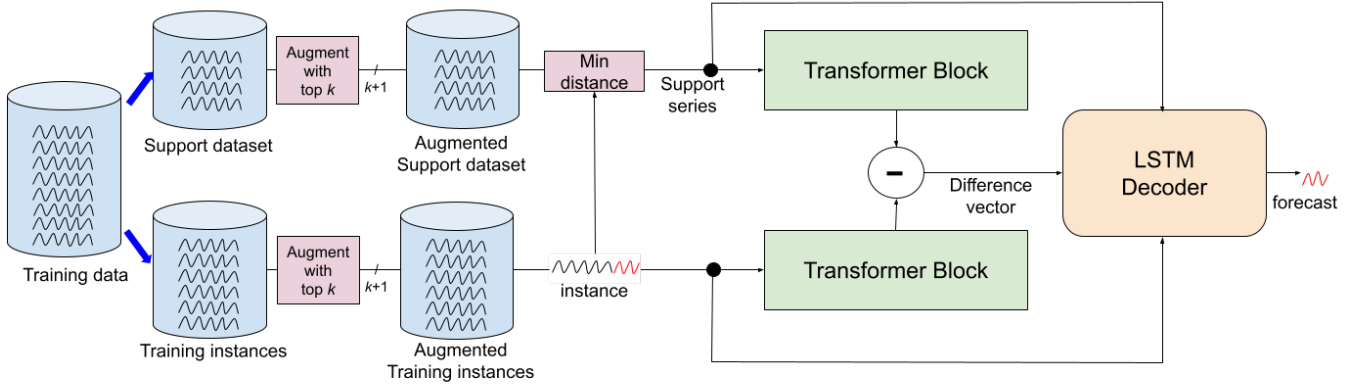


Fig. 1: The proposed architecture, showing the augmentation of training instances using time series from the top- k closest sources and reserving one portion of the training data as a support dataset for use during inference.

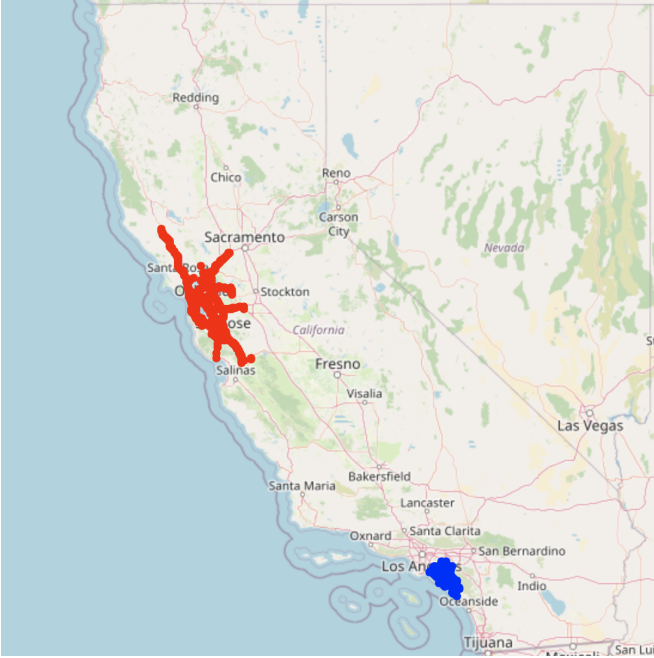


Fig. 2: Map of California showing station locations: red dots indicate test stations (District 4) and blue dots indicate training stations (District 12).

allows us to evaluate the models' ability to perform under zero-shot conditions, assessing their generalization to locations not seen during training.

The training and testing datasets are comprised solely of stations which are a part of a Mainline as determined by Caltrans. Stations with incomplete or missing data were identified and removed from the datasets to ensure data integrity and model reliability. Figure 2 shows the locations of the stations used for training and testing.

3) *Dataset Augmentation*: As this data is generated from physical sensors located on the earth, we will use geographic distance between sensors during the data augmentation stage to convert to multivariate time series. For computational efficiency, geographic distance is approximated using the Haversine distance. Given two locations with latitudes ϕ_1, ϕ_2 and longitudes λ_1, λ_2 , the Haversine distance is

$$a = \sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right),$$

$$d_{\text{hav}} = 2r \arcsin(\sqrt{a}), \quad (3)$$

where r is the radius of the Earth.

4) *Dataset Windowing*: The dataset consists of four-day time series, each containing 192 data points. Each data point represents the flow at a given station over a 30-minute interval. The data points within each four-day period must form a continuous, uninterrupted sequence. We utilize the sliding window method to create the four-day time series, ensuring that the windows are each 4 continuous days starting at midnight on the first day and ending at midnight on the fifth day. We opt for a 30-minute traffic flow forecast to reduce the amount of noise in the dataset.

5) *Dataset Partitioning*: We create two datasets: one containing all of the data from District 12 in 2023, and the other comprising data from the first 3 months of 2023 for District 4. The District 12 dataset is further partitioned into training and validation datasets based on station IDs utilizing a 70/30 split. Each of the three resulting datasets – training, validation, and testing – are then further subdivided into query and support sets using an 85/15 split.

V. RESULTS AND DISCUSSION

Our models were trained and evaluated on an institutional high-performance computing cluster equipped with NVIDIA A100 GPUs. All four models were implemented in PyTorch v2.9.0 with Python 3.11.8. Other libraries used in this work

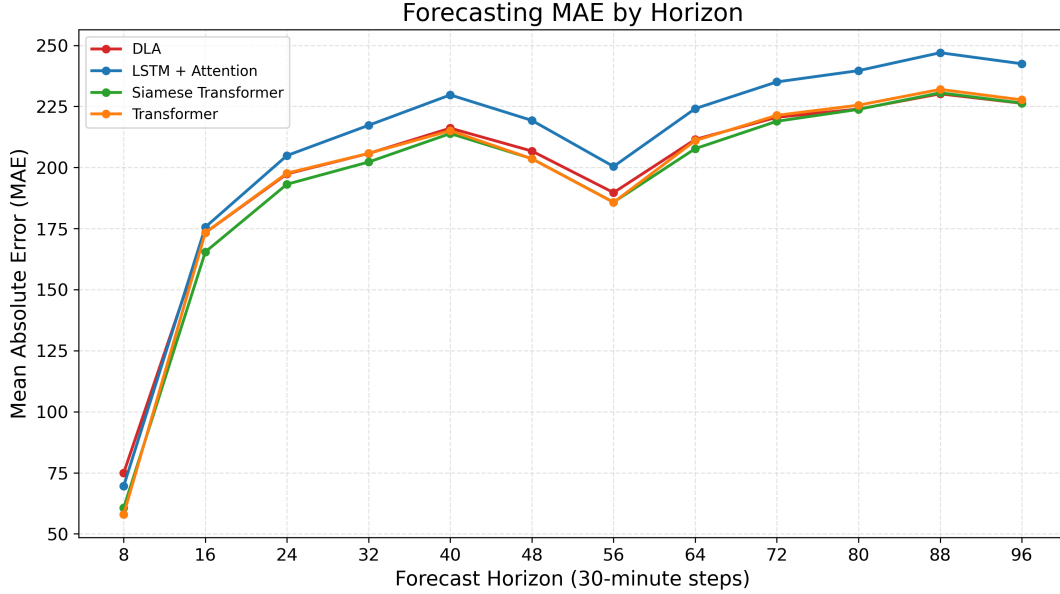


Fig. 3: Mean Absolute Error (MAE) [#Vehicles] across different forecasting horizons for all evaluated models. The x-axis represents the forecasting horizon, measured in 30-minute steps, and the y-axis indicates the corresponding MAE values.

include Pandas v2.3.3, NumPy v2.3.4, Scikit-Learn v1.7.2, SciPy v1.16.2, Matplotlib v3.10.8, and PyYAML v6.0.3.

We evaluated the performance outcomes of the forecasting models using standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Each of these metrics was computed on the PeMS test dataset over 215,062 test samples. Table I provides a comparison of each forecasting model’s performance metrics.

TABLE I: Forecasting Model Performance Comparison on PeMS

Setting	Model	MAE	RMSE	R^2
Multivariate	LSTM*	242.46	328.59	0.8376
Multivariate	DLA	226.29	306.38	0.8613
Multivariate	Siamese Trans.	226.36	309.07	0.8546
Multivariate	Transformer	227.68	312.92	0.8554
Univariate	LSTM*	249.38	335.03	0.8350
Univariate	DLA	238.58	324.66	0.8394
Univariate	Siamese Trans.	231.20	313.43	0.8538
Univariate	Transformer	230.81	316.35	0.8517

* LSTM with Attention mechanism.

MAE and RMSE are reported in the original units of the target sensor measurements. All metrics are computed over 215,062 test samples. Best values within each setting are bolded.

The results show that adding multivariate sensor context improves performance across all model families. For each architecture, the multivariate version achieves lower MAE and RMSE than its univariate counterparts, while also achieving a higher R^2 , indicating that the additional neighboring series provide useful information beyond the target series alone. This pattern suggests that the surrounding sensors capture correlated information that enables the models to resolve local variability and improve the forecast accuracy.

The magnitude of the gain depends on the architecture. The baseline LSTM with Attention improves from a univariate MAE of 249.38 to 242.46 in the multivariate setting, a reduction of approximately 2.77%. DLA reduces the MAE from 238.58 to 226.29, representing a 5.15% reduction, and the largest improvement among the evaluated models. The Siamese transformer architecture also benefits from the multivariate input, reducing the MAE from 231.20 to 226.36, a reduction of approximately 2.09%, while the Transformer reduces the error from 230.81 to 227.68, a reduction of only 1.36%. Similar trends are observed in RMSE and R^2 , with the multivariate versions consistently providing a better overall fit to the target trajectories. These results indicate that multivariate context is beneficial regardless of whether the model is a direct recurrent baseline, a support aware recurrent architecture, or a simple Transformer-based method. Furthermore, the strongest multivariate results are achieved by the support aware models which suggests that combining neighboring sensor context with support conditioned representations yields the most effective forecasting behaviour on PeMS.

Figure 3 displays the MAE across the tested forecasting horizons for the multivariate models. The horizon wise behavior does not follow a simple steady increase; instead, error rises sharply from the short horizons to about 40 steps, where it decreases around 48 to 56 steps, before then increasing again towards the longest horizons. As each step corresponds to 30 minutes, the dip in MAE at 48 steps aligns with the one-day cycle in the traffic data and is consistent with the strong daily seasonality present in the PeMS dataset.

The relative ranking of the models changes with horizon length. The Transformer model attains the lowest MAE at

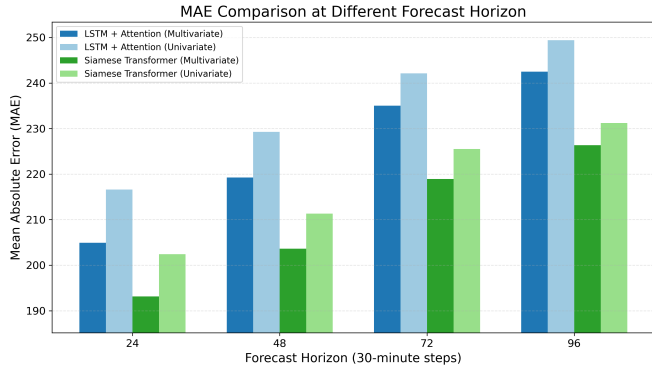


Fig. 4: MAE Comparison at Different Forecast Horizon

the shortest horizon of 8 steps, the Siamese transformer model is strongest across intermediate horizons from 16 to 40 steps and from 64 to 80 steps, and the DLA achieves the best performance at the longest horizons of 88 and 96 steps. The Transformer is also the best at 48 and 56 steps. This pattern suggests that no single architecture dominates at every horizon, but the support-aware models remain the most competitive overall and provide the best performances on the forecasting task (beyond the initial short horizons).

Figure 4 compares the multivariate and univariate versions of the LSTM with Attention and Siamese transformer models at forecasting horizons of 24, 48, 72, and 96 steps. For both model families, the multivariate setting yields lower MAE at every horizon, showing that the added neighboring sensor series provide a consistent benefit from medium to long range forecasting. The largest gains appear at the 24-step horizon, where the multivariate LSTM reduces the error from 216.61 to 204.89 and the multivariate Siamese transformer reduces error from 202.43 to 193.15. Although we observe a narrowing of the gap at longer horizons, the multivariate versions perform better even at 96 steps, thus indicating that the advantage of incorporating contextual sensor information is persistent across the full forecasting window.

VI. CONCLUSIONS

In this work, we extended zero-shot time series forecasting to a multivariate setting by augmenting each target series with related contextual sensor series. Experiments on geographically separated PeMS traffic data showed that multivariate context consistently improved forecasting accuracy over the corresponding univariate version for all evaluated architectures. The strongest overall results were achieved by the support-aware models, indicating that combining contextual sensor information with support-conditioned representations is an effective strategy for zero-shot forecasting. Although no single model performed best at every prediction horizon, support-aware methods remained the most competitive across medium- and long-range forecasting horizons.

These results suggest that zero-shot forecasting benefits not only from transferable temporal representations, but also from carefully selected contextual information from related

sources. Future work should investigate improved support-selection methods, such as learned retrieval, as well as richer approaches for modeling relationships among sensors. It would also be valuable to evaluate these methods on application domains beyond traffic forecasting, such as energy consumption, environmental sensing, healthcare monitoring, or industrial systems, to better understand how well the proposed approach generalizes across different types of multivariate time-series data.

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